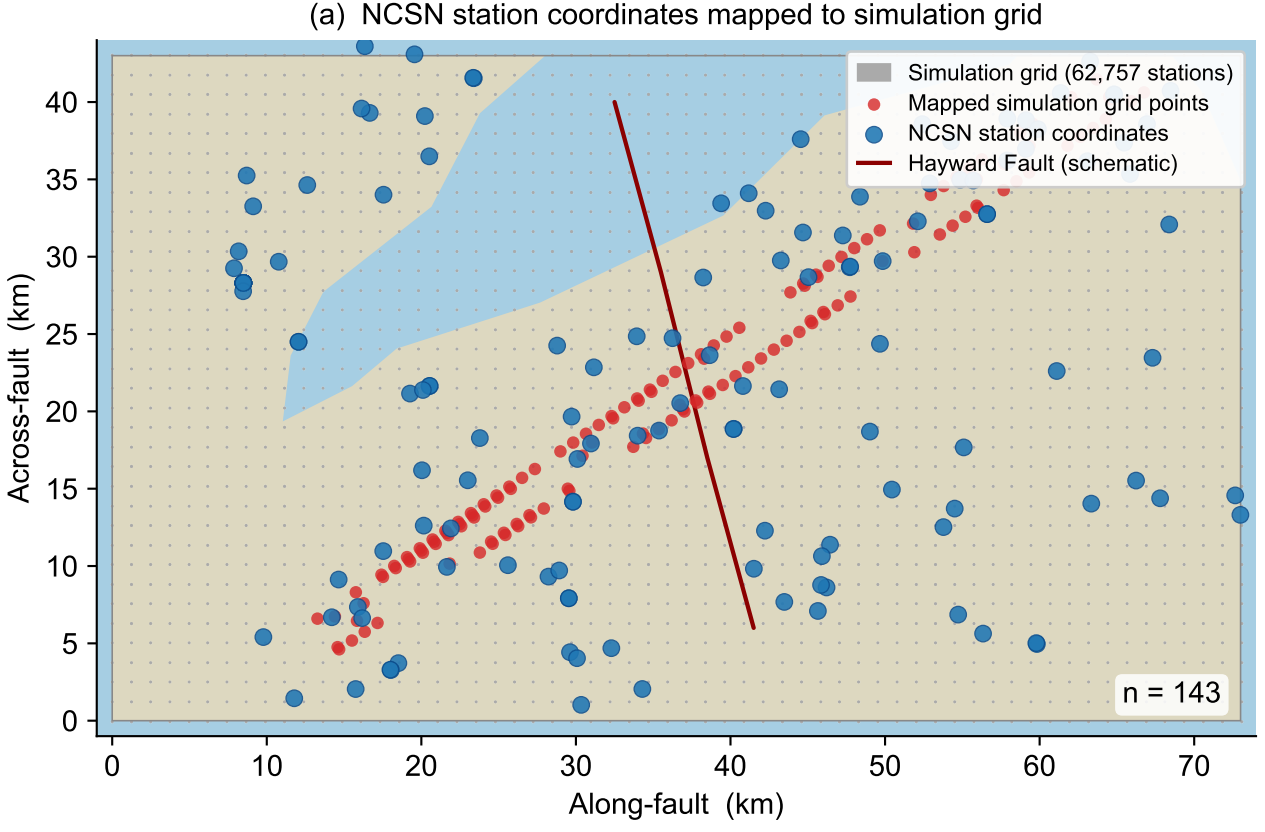
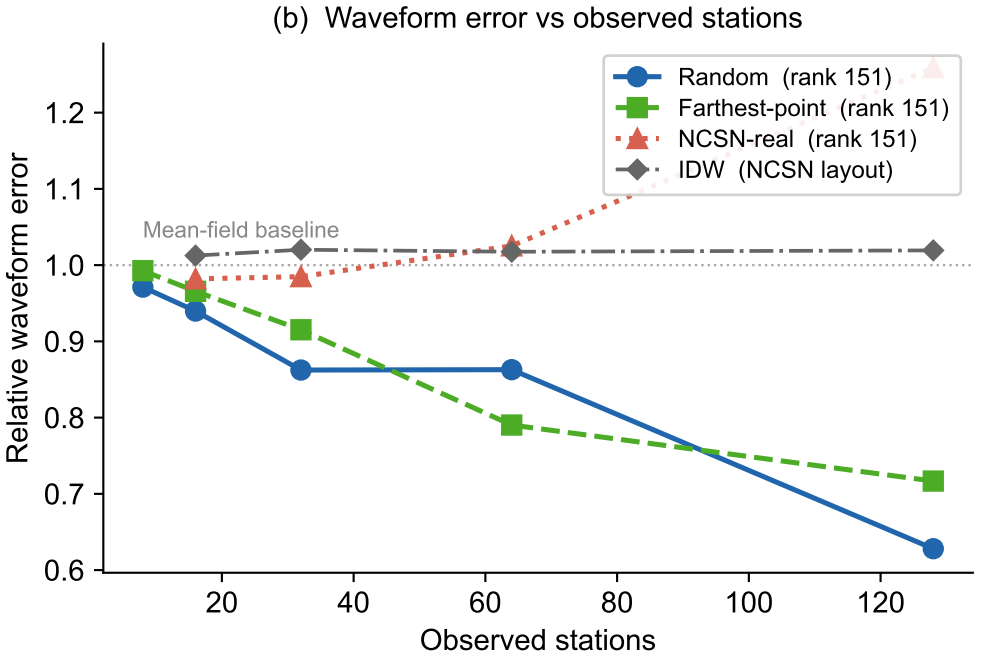
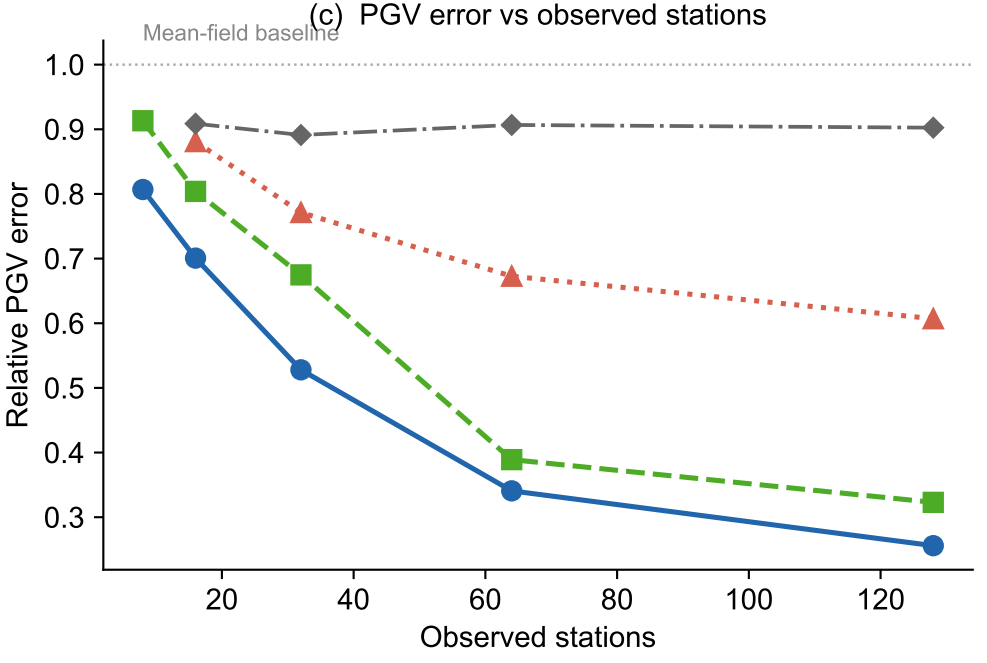

(a) NCSN station coordinates mapped to simulation grid
Simulation grid (62,757 stations)
Mapped simulation grid points
NCSN station coordinates
Hayward Fault (schematic)
n = 143
Across-fault (km)
Along-fault (km)
0
5
10
15
20
25
30
35
40
0
10
20
30
40
50
60
70


(b) Waveform error vs observed stations
Random (rank 151)
Farthest-point (rank 151)
NCSN-real (rank 151)
IDW (NCSN layout)
Mean-field baseline
Relative waveform error
Observed stations
0.6
0.7
0.8
0.9
1.0
1.1
1.2
20
40
60
80
100
120


(c) PGV error vs observed stations
Mean-field baseline
Relative PGV error
Observed stations
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
20
40
60
80
100
120